

DIGITAL SAT — MATH

Theory Companion — Part 1

Algebra & Data Analysis — Theory & Worked Problems

- 13 skills, one representative Hard/Very Hard worked problem per skill
- Algebra: Lines • Systems • Inequalities • Absolute Value • Exponents • Radicals
- Problem-Solving & Data Analysis: Ratios & Proportions • Probability • Unit Conversions • Statistics • Data Displays • Scatterplots • Sampling
- Companion to the full 13-skill, 104-question Math Theory Study Companion

The 13 Skills

1. Lines
2. Systems of Linear Equations
3. Linear Inequalities
4. Absolute Value
5. Exponent Rules
6. Radicals, Rational Exponents & Scientific Notation
7. Ratios & Proportions
8. Probability
9. Unit Conversions & Rates
10. Statistics — Center & Spread
11. Data Displays — Histograms, Dot Plots & Box Plots
12. Scatterplots
13. Sampling, Bias & Margin of Error



PART 2

Problem-Solving & Data Analysis

Ratios & Proportions • Probability • Unit Conversions & Rates • Statistics • Data Displays • Scatterplots • Sampling, Bias & Margin of Error

Core Rules

- Ratio Notation — $a : b = a/b$
- The 'x' Method — If $a:b = m:n$, let $a = mx$ and $b = nx$.
- Proportions & Cross-Multiplication — $a/b = c/d \iff ad = bc$
- Combining Two Ratios — To combine $x:y$ and $y:z$ into $x:z$, scale both ratios so the shared term (y) has the same value in each, using the LCM.
- Direct vs. Inverse Proportion — Direct: $y = kx$. Inverse: $y = k/x$.

Watch Out: Always double-check which quantity is asked for last — many ratio problems ask for one specific part, not the total.

At a bakery in Lyon, the ratio of flour to sugar used in a recipe is 5:3, and the ratio of sugar to butter used is 2:5. If the bakery uses 300 grams of butter for one batch, how many grams of flour does it use?

Formula / Rule Used: Combine two ratios sharing a term by scaling each to a common value for that shared term (using the LCM).

Work through it, then advance for the full step-by-step solution.

Step 1: Write the two given ratios: flour:sugar = 5:3 and sugar:butter = 2:5.

Step 2: The shared term is sugar, which appears as 3 in the first ratio and 2 in the second — find the LCM of 3 and 2, which is 6.

Step 3: Scale flour:sugar = 5:3 by 2 (so sugar becomes 6): flour:sugar = 10:6.

Step 4: Scale sugar:butter = 2:5 by 3 (so sugar becomes 6): sugar:butter = 6:15.

Step 5: Since both ratios now share the same sugar value, combine them into one three-part ratio: flour:sugar:butter = 10:6:15.

Step 6: The butter amount, 300 g, corresponds to the 15 parts, so one part = $300/15 = 20$ g.

Step 7: Find the flour amount: flour = $10 \times 20 = 200$ grams.

Answer: 200 grams of flour.

Tip: When two ratios share a term, scale each ratio (using the LCM of the shared term's two values) so that term matches in both, then merge them into one three-part ratio before using any given total.

Trap: Because sugar is '3' in the first ratio and '2' in the second, a rushed student might just concatenate the numbers as flour:sugar:butter = 5:3:5 (reusing the sugar:butter figures unscaled) instead of finding the LCM of 3 and 2 and scaling both ratios to a common sugar value of 6, producing an incorrect combined ratio and a wrong flour amount.

Tempting Wrong Answers — and Exactly What Went Wrong

300 grams

Concatenated the two ratios directly as flour:sugar:butter = 5:3:5 (reusing butter's raw ratio number without scaling to match sugar's LCM value of 6), overstating the flour amount.

500 grams

Applied the flour:sugar ratio (5:3) directly to the butter amount as if sugar and butter were interchangeable, computing flour = $300 \times 5/3$ instead of combining through the properly scaled three-part ratio.

120 grams

Correctly found one part = $300/15 = 20$ g but then multiplied by sugar's part count (6) instead of flour's (10), reporting the sugar amount instead of the flour amount.

Shortcut: Since only the flour amount is needed, you can isolate the flour:butter portion of the combined ratio (10:15) and solve $300 \times 10/15 = 200$ directly, skipping the sugar amount entirely.

Desmos Tip: Type $300 \times (10/15)$ into the calculator to instantly confirm the flour amount is 200 grams.

Core Rules

- Basic Probability — $P(\text{event}) = \text{favorable outcomes} / \text{total outcomes}$
- Complement — $P(A') = 1 - P(A)$
- 'And' — Independent vs. Dependent — Independent: $P(A \text{ and } B) = P(A) \times P(B)$. Dependent: $P(A \text{ and } B) = P(A) \times P(B|A)$.
- 'Or' — Mutually Exclusive vs. Not — Mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$. General: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.
- Conditional Probability — $P(A|B) = P(A \text{ and } B) / P(B)$

Watch Out: 'Without replacement' changes the denominator (and sometimes the numerator) for every subsequent draw — always recount what remains.

At the Port of Piraeus, random safety inspections are performed on shipping containers. Historical records show that, independent of any other container, 1 out of every 10 containers fails its safety check. The port handles about 4,000 containers per month. If an inspector randomly selects 3 containers to check, what is the probability that at least one of the 3 fails its safety check?

Formula / Rule Used: Complement rule for 'at least one': $P(\text{at least one}) = 1 - P(\text{none})$.

Work through it, then advance for the full step-by-step solution.

Step 1: Translate the rate into a probability: 1 out of every 10 containers fails, so $P(\text{fail}) = 1/10$ and $P(\text{pass}) = 1 - 1/10 = 9/10$.

Step 2: The monthly total of 4,000 containers is not needed -- the question only concerns the 3 selected containers.

Step 3: 'At least one fails' is the complement of 'none of the 3 fail,' so find $P(\text{none fail})$ first.

Step 4: Since the containers are independent, multiply: $P(\text{none fail}) = 9/10 \times 9/10 \times 9/10 = 729/1000$.

Step 5: Apply the complement rule: $P(\text{at least one fails}) = 1 - 729/1000$.

Step 6: Subtract: $1000/1000 - 729/1000 = 271/1000$.

Answer: $271/1000 (= 0.271)$

Tip: Whenever a problem asks for the probability of 'at least one' success among repeated independent trials, immediately switch to finding the probability of 'none' and subtract from 1 -- it's almost always faster than adding up every possible case.

Trap: A rushed student multiplies the single-event probability by the number of trials -- $3 \times 1/10 = 3/10$ -- treating the probabilities as if they simply add, but that overcounts cases where more than one container fails and ignores how the complement actually compounds.

Tempting Wrong Answers — and Exactly What Went Wrong

3/10

Multiplied the single-container failure probability by the number of trials ($3 \times 1/10$) instead of applying the complement rule, treating probabilities as additive across independent trials.

729/1000

Correctly computed $P(\text{none fail}) = 9/10 \times 9/10 \times 9/10$ but forgot the final step of subtracting from 1, submitting the complement of the requested event instead of the event itself.

243/1000

Computed $P(\text{exactly one fails}) = 3 \cdot (\frac{1}{10}) \cdot (\frac{9}{10})^2 = \frac{243}{1000}$, mistaking 'at least one' for 'exactly one' fails and only counting the single-failure case.

Shortcut: The complement approach IS the shortcut here -- directly listing every combination of 1, 2, or 3 failing containers would take three separate calculations instead of one subtraction.

Desmos Tip: Type $1 - (\frac{1}{10})^3$ directly into the calculator to get 0.271 instantly, avoiding any arithmetic slip when cubing $\frac{1}{10}$ by hand.

Core Rules

- Unit Conversion Method — $\text{Given} \times (\text{desired unit} / \text{given unit}) = \text{result}$
- Multi-Step Conversions — Chain multiple conversion factors in a row when no direct factor is given (e.g. seconds $\rightarrow$ minutes $\rightarrow$ hours).
- Squared & Cubed Units — $1 \text{ unit}^2 = \text{conversion factor}^2$; $1 \text{ unit}^3 = \text{conversion factor}^3$
- Density & Rate — Density = mass / volume. Rate = quantity / time.

Watch Out: Converting from a smaller unit to a larger one means dividing; larger to smaller means multiplying — check the direction before you calculate.

A speed camera near Lisbon reports a car's speed in two different units at the same instant: feet per second and miles per hour. The technician notices the feet-per-second reading is exactly 14 more than the miles-per-hour reading. Given that 1 mile = 5280 feet and 1 hour = 3600 seconds, what is the car's speed in miles per hour?

Formula / Rule Used: Convert one variable's units using the exact conversion fraction, then solve the resulting linear equation.

Work through it, then advance for the full step-by-step solution.

Step 1: Let v = the car's speed in miles per hour.

Step 2: Convert v to feet per second: $v \text{ mph} = v \times (5280 \text{ ft} / 3600 \text{ s}) = (22/15)v \text{ ft/s}$.

Step 3: Translate the given relationship into an equation: the ft/s reading equals the mph reading plus 14, so $(22/15)v = v + 14$.

Step 4: Clear the fraction by multiplying every term by 15: $22v = 15v + 210$.

Step 5: Subtract $15v$ from both sides: $7v = 210$.

Step 6: Divide by 7: $v = 30$.

Step 7: Check: $30 \text{ mph} = 30 \times 22/15 = 44 \text{ ft/s}$, and $44 - 30 = 14$, which matches the given condition.

Answer: 30 mph

Tip: When a problem relates the same speed expressed in two different units, convert everything into one variable using the exact conversion factor as a fraction (don't round it to a decimal), then solve algebraically.

Trap: A student who rounds $5280/3600$ to a decimal too early, or who writes the equation backwards as $v - (22/15)v = 14$ (assuming the mph reading is the larger one), will get an incorrect or nonsensical value instead of correctly treating the ft/s reading as the larger one.

Tempting Wrong Answers — and Exactly What Went Wrong

28 mph

Rounded the conversion factor $5280/3600$ to 1.5 instead of using the exact fraction $22/15$, then solved $1.5v - v = 14$ to get $v = 28$.

44 mph

Correctly solved for the ft/s value (44) but reported that number as the final mph answer instead of continuing to identify v itself.

≈ 6.5 mph

When isolating v from $7v = 210$ (or the equivalent $(7/15)v = 14$), multiplied both sides by $7/15$ instead of dividing by it (multiplying by the reciprocal $15/7$), inverting the fraction incorrectly.

Shortcut: Keep the conversion factor as the exact fraction $22/15$ and immediately clear denominators by multiplying the whole equation by 15 — this turns it into $22v = 15v + 210$ with no decimals or rounding anywhere.

Desmos Tip: Graph $y = (22/15)x$ and $y = x + 14$ and find their intersection point — the x-coordinate (30) is the mph speed and the y-coordinate (44) confirms the ft/s reading.

Core Rules

- Mean, Median, Mode, Range — Mean = sum/n . Median = middle value (average the two middle values if n is even). Mode = most frequent. Range = $\text{max} - \text{min}$.
- Weighted Average — Weighted avg = $(\text{mean1} \times n1 + \text{mean2} \times n2 + \dots) / (n1 + n2 + \dots)$
- Standard Deviation (conceptual only) — You are never asked to calculate it by hand — only to compare: a more spread-out data set has a higher standard deviation, regardless of the actual values.
- Effect of Outliers — An outlier pulls the mean toward itself but barely affects the median; it also inflates both the range and the standard deviation.
- Mean vs. Median & Skew — Right-skew: $\text{mean} > \text{median}$. Left-skew: $\text{mean} < \text{median}$. Symmetric: $\text{mean} \approx \text{median}$.

Watch Out: When a question removes or adds one value and asks for a new mean, use $\text{Sum} = \text{Mean} \times n$ to recover the total — it's almost always faster than recomputing from scratch.

A math club in Vienna gives a 150-point placement test. Six students have already been scored, from lowest to highest: 62, 74, 81, 85, 90, 96. A seventh student, whose score is higher than all six of these, is added to the group. Once the seventh score is included, the mean of all seven scores equals the median of all seven scores plus 2. What is the seventh student's score?

Formula / Rule Used: Median of an odd-sized list is the single middle value once sorted; mean = sum / count.

Work through it, then advance for the full step-by-step solution.

Step 1: Since the new score is higher than all six existing scores, the sorted list of seven scores is 62, 74, 81, 85, 90, 96, x.

Step 2: With 7 values, the median is the 4th value in the sorted list, which is 85 -- this stays 85 no matter how large x is, as long as x remains the largest.

Step 3: Set up the mean: $(62+74+81+85+90+96+x)/7 = (488+x)/7$.

Step 4: The condition says mean = median + 2, so $(488+x)/7 = 85 + 2 = 87$.

Step 5: Multiply both sides by 7: $488 + x = 609$.

Step 6: Solve for x: $x = 609 - 488 = 121$.

Answer: 121 points

Tip: When a data set has an odd number of values and a new value is added only at the very top or bottom, find the median first -- it often doesn't move at all, which simplifies the rest of the problem.

Trap: Treating the 7-score set like an even-sized list and averaging the 3rd and 4th values (81 and 85) instead of taking the single 4th value (85) is exactly the mistake this problem is built to catch.



Tempting Wrong Answers — and Exactly What Went Wrong

107

Treated the 7-value set as if it were even-sized, averaging the 3rd and 4th values (81 and 85) to get a 'median' of 83, then solved $(488+x)/7 = 85$ instead of 87.

34

Used the correct median (85) and required mean (87), but divided by 6 instead of 7, forgetting the new score makes this a seven-value set, not six.

609

Correctly computed $7 \times 87 = 609$ but reported this total sum as the final answer instead of subtracting 488 to isolate x .

Shortcut: Since the median (85) is fixed and unaffected by x , you can jump straight to mean = 87 and solve the one-line equation $488+x=609$ without extra setup.

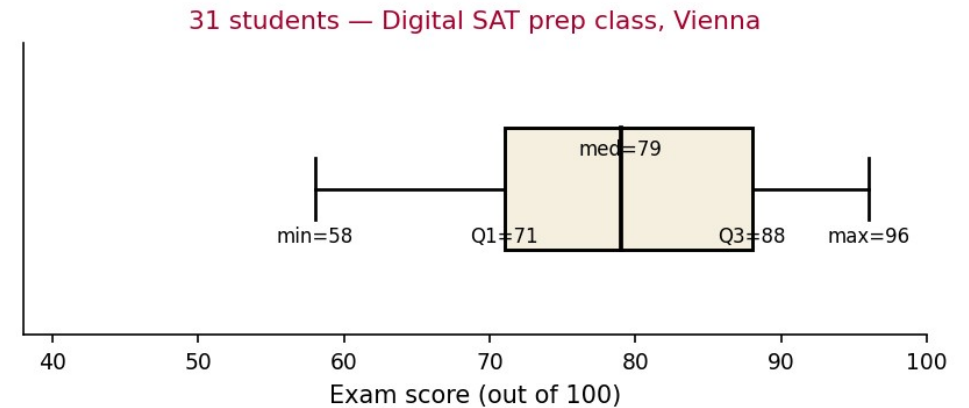
Desmos Tip: Type `mean({62,74,81,85,90,96,121})` into the calculator to confirm it equals 87, and `median(...)` of the same list to confirm 85.

Core Rules

- Histograms — Continuous numerical data grouped into intervals with NO gaps between bars; the shape (bell, right-skewed, left-skewed) reveals the distribution's symmetry.
- Dot Plots — Each dot is one data point above a number line — the exact frequency, clusters, gaps, and outliers are all directly visible, unlike a histogram.
- Box Plots (Five-Number Summary) — Min, Q1, Median, Q3, Max
- Reading the Median from a Dot Plot — Cross out the highest and lowest remaining values one pair at a time until one (or two, to average) value remains in the middle.

Watch Out: In a box plot, a longer whisker or box section on one side signals skew in that direction — you can answer many box-plot questions without ever knowing the raw data.

A box plot summarizes the exam scores (out of 100) of 31 students in a Digital SAT prep class in Vienna: minimum = 58, Q1 = 71, median = 79, Q3 = 88, maximum = 96. A 32nd student transfers in with a score of 47. Using the standard rule that a value is an outlier if it falls more than $1.5 \times \text{IQR}$ below Q1 or above Q3, find the lower fence and determine whether the new student's score is an outlier.



Formula / Rule Used: $\text{IQR} = Q3 - Q1$; outlier fences = $Q1 - 1.5(\text{IQR})$ and $Q3 + 1.5(\text{IQR})$.

Work through it, then advance for the full step-by-step solution.

Step 1: Find the IQR from the box plot: $IQR = Q3 - Q1 = 88 - 71 = 17$.

Step 2: Multiply by 1.5: $1.5 \times 17 = 25.5$.

Step 3: Find the lower fence: $\text{lower fence} = Q1 - 25.5 = 71 - 25.5 = 45.5$.

Step 4: Compare the new student's score to the lower fence: 47 vs. 45.5.

Step 5: Since $47 > 45.5$, the score of 47 lies inside the fence, not beyond it.

Step 6: Conclude: the new student's score of 47 is NOT an outlier by the $1.5 \times IQR$ rule, even though it is well below the original minimum of 58.

Answer: Lower fence = 45.5; 47 is NOT an outlier ($47 > 45.5$).

Tip: To test for an outlier, always compute the actual fence ($Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$) before comparing — never judge 'extreme-ness' just by comparing to the box plot's own minimum or maximum.

Trap: Because 47 is lower than the original minimum of 58, it looks obviously extreme — but the fence calculation shows 47 still falls inside the allowed range, so the intuitive answer (outlier) is wrong.

Tempting Wrong Answers — and Exactly What Went Wrong

Outlier (fence = 54)

Computed the fence as $Q1 - IQR$ ($71 - 17 = 54$) instead of $Q1 - 1.5 \times IQR$, forgetting the 1.5 multiplier entirely.

Outlier, since 47 is below the original minimum of 58

Assumed any new value below the group's previous minimum must automatically be an outlier, without computing a fence at all.

Outlier (fence = 53.5)

Used the median (79) instead of $Q1$ (71) as the base for the lower fence, computing $79 - 25.5 = 53.5$ and comparing 47 to that.

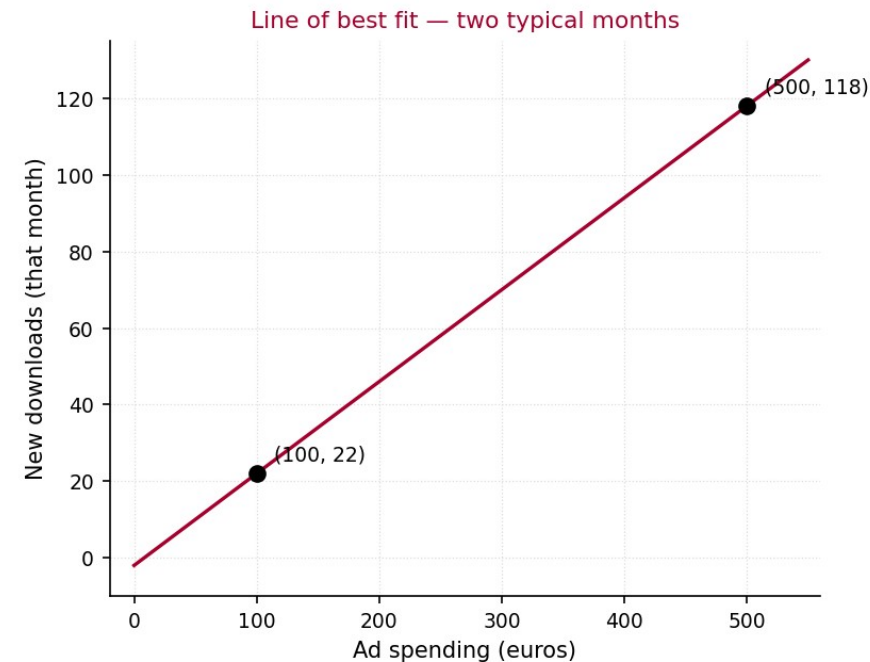
Shortcut: Once you know $Q1$ and IQR , compute the fence in one line — $Q1 - 1.5 \times IQR$ — and just check whether 47 clears it; no need to re-sort or re-derive the whole distribution.

Core Rules

- Line of Best Fit — $y = mx + b$ (linear case)
- Residuals — Residual = actual value - predicted value
- Interpreting Slope & Intercept in Context — Slope = predicted change in y per one-unit increase in x . Y-intercept = predicted value of y when $x = 0$ — sanity-check whether that's realistic in context.
- Types of Association — Positive: as x increases, y increases. Negative: as x increases, y decreases. No association: no clear pattern.
- Nonlinear Models — Quadratic: $y = ax^2 + bx + c$. Exponential: $y = a \cdot b^x$.

Watch Out: To find two points ON the line of best fit (not the data points themselves), read directly off the line — data points scattered near it are not the same as points on it.

A mobile-game studio in Rotterdam ran a monthly analysis linking its social-media advertising spending (in euros) to the number of new app downloads that month. Studying 14 months of data, analysts noted that a month with 100 euros of ad spending averaged 22 new downloads, while a month with 500 euros of ad spending averaged 118 new downloads. Using these two typical months as points on the line of best fit, (a) find the predicted number of new downloads for a month with 350 euros of spending, and (b) determine whether the model's y-intercept is realistic in context.



Formula / Rule Used: Slope-intercept equation from two data points; realism check on the y-intercept.

Work through it, then advance for the full step-by-step solution.

Step 1: Translate the two described months into ordered pairs: (100, 22) and (500, 118), where x = euros spent and y = new downloads.

Step 2: Find the slope: $m = (118 - 22)/(500 - 100) = 96/400 = 0.24$ downloads per euro.

Step 3: Use point-slope form with (100, 22): $22 = 0.24(100) + b$, so $22 = 24 + b$.

Step 4: Solve for b : $b = 22 - 24 = -2$.

Step 5: Write the line of best fit: $y = 0.24x - 2$.

Step 6: Predict downloads for $x = 350$: $y = 0.24(350) - 2 = 84 - 2 = 82$ downloads.

Step 7: Check the y -intercept: at $x = 0$ the model predicts $y = -2$ downloads, which is impossible since a download count cannot be negative, so the intercept is not realistic — it is just an artifact of extending the line past the observed data.

Answer: 82 downloads predicted at 350 euros; the y -intercept of -2 is not realistic because download counts cannot be negative.

Tip: When a word problem describes two 'typical' months or cases instead of giving a table, translate them into ordered pairs first, then treat the problem exactly like finding a line through two points.

Trap: A rushed student might treat 22 and 118 as slope values instead of y -coordinates paired with their given x -values, or accept -2 downloads as a valid prediction without questioning whether a negative count makes sense.

Tempting Wrong Answers — and Exactly What Went Wrong

≈1064 downloads predicted

Flipped the slope ratio, computing $(500-100)/(118-22) = 400/96$ instead of $96/400$, giving an inflated slope and a wildly overestimated prediction at $x = 350$.

106 downloads predicted

Set $b = 22$ directly from the point $(100, 22)$ without solving $22 = 24 + b$ for b , effectively treating the y -coordinate itself as the intercept and using $y = 0.24x + 22$.

82 downloads predicted, with the y -intercept of -2 accepted as realistic

Computed the prediction correctly but didn't check whether a negative predicted download count at $x = 0$ makes sense, accepting -2 as plausible instead of recognizing counts can't be negative.

Shortcut: Since the two x -values are 400 apart, the slope $96/400 = 0.24$ can be found in one line without formally writing point-slope form first.

Desmos Tip: Plot $(100, 22)$ and $(500, 118)$ in Desmos and use its built-in linear regression on the two points (or graph $y = 0.24x - 2$) to confirm the line crosses below $y = 0$ near $x = 0$.

Core Rules

- Population vs. Sample — A reliable sample must be both large AND randomly selected from the population you actually want to draw conclusions about.
- Types of Bias — Sampling bias: the population itself wasn't randomly selected. Self-selection bias: participants volunteered. Nonresponse bias: certain groups didn't respond. A larger sample never fixes a biased selection method.
- Margin of Error — True population value $\approx$ sample value $\pm$ margin of error
- Drawing Valid Conclusions — Results generalize ONLY to the exact population that was randomly sampled. Correlation is never proof of causation — only a controlled, randomized experiment supports a cause-and-effect claim.

Watch Out: When comparing two studies, a smaller sample size (all else equal) means a LARGER margin of error — this trips up students who assume 'smaller number' always means more precise.

The city council of Rotterdam wants to know what fraction of all 620,000 residents support a proposed bike-lane expansion. The council posts a link to an online survey on the city's official social media page and invites anyone who sees the post to click and respond. The survey stays open for two weeks and receives 1,850 responses — more than double the roughly 800 responses the council's mailed paper surveys usually generate. Of the 1,850 online respondents, 74% support the expansion. A council member wants to report that '74% of Rotterdam residents support the bike-lane expansion.' What single flaw in the sampling method makes this specific conclusion invalid, regardless of how many responses were collected?

Formula / Rule Used: Identifying flaws in a study design: check whether participants were randomly selected or self-selected before trusting any generalization to the population.

Work through it, then advance for the full step-by-step solution.

Step 1: Note that the sample was NOT drawn randomly from all 620,000 residents — whether someone is included depended entirely on whether they saw the post and chose to click it.

Step 2: Consider whether the sample size (1,850) is the issue — it isn't: 1,850 is actually a fairly large sample, and comparing it to the 800 mailed responses is irrelevant to whether the method itself is unbiased.

Step 3: Consider whether the two-week window caused the problem — timing isn't the flaw either; even a random sample collected over two weeks would still be valid.

Step 4: Recognize the real issue: this is a voluntary response sample. People with strong opinions (often more motivated to click and respond) are systematically over-represented compared to indifferent or opposed residents.

Step 5: Because respondents chose themselves rather than being randomly selected, the 74% reflects motivated social-media users, not a representative cross-section of all 620,000 residents.

Answer: Voluntary response (self-selection) bias: only residents who chose to click the online survey are represented, so the 74% cannot be generalized to all Rotterdam residents.

Tip: When a study lets participants decide for themselves whether to respond (a posted link, a call-in poll, a mail-back form), check for self-selection bias first — no sample size, however large, fixes a non-random selection method.

Trap: The '1,850 vs. 800' comparison is designed to tempt you into thinking a bigger sample fixed the problem — sample size only affects margin of error, not whether the sample was randomly selected in the first place.

Tempting Wrong Answers — and Exactly What Went Wrong

The sample size (1,850) is too small to be reliable

Confuses margin-of-error concerns (which shrink as n grows) with selection bias; a bigger sample only makes a biased method's estimate more precisely wrong, not more valid.

The two-week survey window is too short to capture opinions

Focuses on an irrelevant detail (duration) instead of recognizing that no survey window length fixes a fundamentally non-random selection process.

The survey undercounts residents without internet access

Names a real but secondary issue (undercoverage) instead of the defining flaw — even if every resident had internet access, letting people self-select into responding would still bias the result.

Shortcut: As soon as you see 'anyone who wants to can click and respond,' flag voluntary response bias immediately — there's no need to evaluate sample size, timing, or response rate at all.